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Business Intelligence Theory Assignment.

Q1. What is the association rule mining? Explain the terms support, confidence?

→ Associate rule mining is a fundamental technique in data mining used to discover interesting relationships, patterns, or associations among variables in large datasets.

- It is widely applied in domains such as market basket analysis, recommendation systems, healthcare analytics and web usage mining.

- The primary goal is to identify rules of form $X \rightarrow Y$, where X and Y are sets of items, indicating that the occurrence of X in a transaction is likely to be followed by occurrence of Y .

- This technique helps businesses understand customer behaviour, such as which products are frequently purchased, enabling them to make informed decisions because of promotion and cross selling strategies.

- For eg - supermarket dataset is an association -
Bread, Butter $\rightarrow$ Milk.

Key terms -

1) Support - Support is a measure that indicates how frequently a particular itemset appears in the dataset. It reflects the popularity or occurrence of an item or item combination within all transactions. In simple terms support tells us how common a rule is in the entire dataset.

- If a rule has high support, it means the combination of items appears frequently in the dataset, making it more relevant for analysis.
- Low support indicates rule occurs rarely and may not be significant.

2) Confidence -

Confidence is a measure that indicates the strength or reliability of an association rule. It shows how often items in Y appear in transactions that already contain X. In other words it measures probability that Y occurs given that X has occurred.

$$\text{Confidence } (X \rightarrow Y) = \frac{\text{No. of transactions } XY}{\text{No. of transaction with } X.}$$

- High confidence means that when X occurs, Y is likely to occur as well. making the rule strong and useful. Low confidence suggests a weak relationship in X and Y.

Q2. What is the difference between hierarchical clustering and partitioning?

→ Hierarchical clustering	Partitioning.
1. Builds a hierarchy (tree like structure of clusters)	1. Divides data into a fixed number of clusters.
2. Does not require number of clusters in advance (in most cases)	2. Requires number of clusters to be specified beforehand.
3. Produces a dendrogram (tree diagram)	3. Does not produce a dendrogram.
4. Can be agglomerative (bottom up) or divisive (top down)	4. Usually uses iterative relocation of points (eg k means).
5. Once a merge split is done, it cannot be undone.	5. Data points can be reassigned during iterations.
6. Computationally expensive slow for large dataset.	6. Faster and efficient for large dataset.
7. Provides better insight into data structure	7. Focuses on optimizing cluster centers.

4.

Q3. Explain logistic regression with example -

→ • Logistic regression is a supervised learning classification algorithm used to predict categorical outcomes, typically binary (such as yes/no, 0/1)

• Unlike linear regression which predicts continuous values, logistic regression predicts the probability of a class and assigns a label based on that probability.

• Sigmoid function -

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

• The output of the sigmoid function lies between 0 and 1.
• If output is greater than 0.5 the instance is 1 otherwise 0.

• Working -

1. Input - features are taken (X variables).
2. A linear combination is formed.
3. The result is passed through the sigmoid function.
4. A probability value is obtained.
5. Based on threshold (usually 0.5) the output is classified.

Example - Predict whether order is delayed or not.

Input features :

- Delivery days.
- Freight cost.

Output:

- 1 $\rightarrow$ Delayed.
- 0 $\rightarrow$ On time.

Sample data -

<u>Delivery data</u>	<u>Freight</u>	<u>Output</u>
3	50	0
10	200	1
7	120	1

If model produces probability of 0.8 then,
Since $0.8 > 0.5$ it is classified as Delayed.

• Advantages -

- Simple and easy to implement.
- Efficient for binary classification problems.
- Provides probabilistic output.

Limitation -

- It assumes a linear relationship between input variables and log odds.
- Not suitable for highly complex or non linear data.

Q4. Explain BI application in CRM -

→ • Business Intelligence in customer relationship management helps organizations analyze customer data to improve relationships, sales, customer satisfaction.

• Applications of BI in CRM.

1) Customer segmentation -

BI helps divide customers to groups based on behaviour, demographics and purchasing patterns. This enables targeted marketing.

2) Customer retention analysis -

Identifies customers who are likely to leave churn and helps businesses take preventive actions.

3) Sales analysis -

Tracks customer purchases and sales trends to improve sales strategies and forecasting.

4) Personalized marketing -

BI enables personalized recommendations and offers based on customer preferences.

5) Customer lifetime value -

Helps estimate the long term value of a customer, allowing businesses to focus on high value customers.

Q5) Define business intelligence and Explain any 3 tools - for business.

◦ → Business intelligence BI refers to technologies, applications and practices used to collect, analyze and present business data to support decision making.

◦ Tools for BI -

1. Power BI -

- Developed by Microsoft.
- Provides interactive dashboards and reports.
- Connects to multiple data sources.
- Used for data visualization and analysis.

2. Tableau -

- Powerful data visualization tool.
- Easy drag and drop interface.
- Creates interactive charts and dashboards.
- Widely used in business analytics.

3. Qlick view -

- Uses in memory data processing.
- Fast data analysis.
- Provides associative data modeling.
- Useful for discovering hidden insights.

Q6.

Q6. Explain the role of BI in finance and Marketing.

→ • Role of BI in finance -

1. Financial reporting - generates accurate real time financial reports.
2. Budgeting and forecasting - Helps predict future financial performance using historical data.
3. Risk management - Identifies financial risks and supports decision making.
4. Fraud detection - detects unusual transactions and suspicious activities.

• Role of BI in marketing -

1. Market analysis - Analyses customer trends and market conditions.
2. Campaign performance - Evaluates effectiveness of marketing campaigns.
3. Customer behaviour analysis - Understands buying patterns and preferences.
4. Sales optimization - Improves pricing, promotions and product strategies.